

Chapter 13 - PSG and AY-3-8910

The PSG (Programmable Sound Generator) is the General Instrument AY-3-8910 and its Yamaha-relabelled YM-2149 sibling: a three-voice square-wave chip with a noise generator, an envelope generator, and two general-purpose I/O ports. Intuition Engine's PSG is faithful to both, so existing ZX Spectrum 128, Amstrad CPC, MSX, and Atari ST PSG music plays unchanged.

13.1 What the PSG can show

Item	Value
Tone channels	3 (A, B, C)
Noise generator	1, shared between channels via mixer
Envelope	1, shared between channels via control
Volume control	Per-channel 4-bit volume
Register count	16 (0x00–0x0F)

13.2 The register block

The PSG sits at 0xF0C00–0xF0C0F. Each register is one byte at a 4-byte-stride MMIO slot. The standard AY-3-8910 register layout is preserved:

Address	Reg	Name
0xF0C00	0	Channel A frequency, low byte
0xF0C01	1	Channel A frequency, high nibble
0xF0C02	2	Channel B frequency, low byte
0xF0C03	3	Channel B frequency, high nibble
0xF0C04	4	Channel C frequency, low byte
0xF0C05	5	Channel C frequency, high nibble
0xF0C06	6	Noise period, 5 bits
0xF0C07	7	Mixer / I/O control
0xF0C08	8	Channel A level
0xF0C09	9	Channel B level
0xF0C0A	10	Channel C level
0xF0C0B	11	Envelope period, low byte
0xF0C0C	12	Envelope period, high byte
0xF0C0D	13	Envelope shape (4 bits)
0xF0C0E	14	I/O port A
0xF0C0F	15	I/O port B

13.2.1 Channel frequency

A channel's pitch is set by a 12-bit divider: bits 0–7 from the low register and bits 8–11 from the low nibble of the high register. The output frequency is

$$f = \text{clock} / (16 \times \text{divider})$$

with the AY-style clock typically chosen from the machine being reproduced. The IE ships these clock constants for fidelity:

Clock	Value (Hz)
PSG_CLOCK_ATARI_ST	2,000,000
PSG_CLOCK_ZX_SPECTRUM	1,773,400
PSG_CLOCK_CPC	1,000,000
PSG_CLOCK_MSX	1,789,773

13.2.2 Mixer (register 7)

Bit	Field	Meaning
0	~TONE_A	0 = tone A active.
1	~TONE_B	0 = tone B active.
2	~TONE_C	0 = tone C active.
3	~NOISE_A	0 = noise routed to A.
4	~NOISE_B	0 = noise routed to B.
5	~NOISE_C	0 = noise routed to C.
6	IO_A_OUT	1 = I/O port A is output.
7	IO_B_OUT	1 = I/O port B is output.

Bits 0–5 are inverted by convention: writing 0 enables the voice.

13.2.3 Level (registers 8–10)

Bit	Field	Meaning
0–3	Level	Volume 0–15.
4	ENV_EN	1 = use envelope; level bits ignored.

13.2.4 Envelope shape (register 13)

Value	Shape
0–3	Decay, hold low.
4–7	Attack, hold low.
8	Saw down (repeat).
9	Decay, hold low (one-shot).

Value	Shape
10	Triangle (alternating).
11	Decay, hold high.
12	Saw up (repeat).
13	Attack, hold high.
14	Triangle (alternating, inverted).
15	Attack, hold low.

Writing to register 13 restarts the envelope, so the same shape value is the conventional retrigger mechanism.

13.3 The PSG+ extension

Bit 0 of the IE-specific PSG_PLUS_CTRL register at 0xF0C20 enables **PSG+** mode. When PSG+ is on, the chip widens its level registers from 4 bits to 8 bits and offers per-channel pan; when off, the chip behaves exactly like an AY-3-8910.

13.4 Hardware-mode AY playback

The PSG also has a small player that streams a register-frame log straight into the chip. This is the format used by Vortex Tracker and most ZX Spectrum 128 music players.

Address	Name	Purpose
0xF0C10	PSG_PLAY_PTR	Address of the register-frame stream.
0xF0C14	PSG_PLAY_LEN	Stream length in bytes.
0xF0C18	PSG_PLAY_CTRL	1 = play, 0 = stop.
0xF0C1C	PSG_PLAY_STATUS	Read-only status.

Each PSG_REG_COUNT (16) bytes in the stream is one PSG frame. The chip applies frames at the rate of the music being played (50 Hz for ZX Spectrum, 60 Hz for NTSC titles, etc.).

13.5 BASIC keywords

Form	Effect
PSGch, freq, vol	Set channel <i>ch</i> 's 8-bit frequency divider and 4-bit volume.
PSG MIXERvalue	Write the mixer/I/O byte (register 7).
PSG ENVELOPEshape, period	Set the envelope shape and 16-bit period.
PSG PLUS ON / PSG PLUS OFF	Enable / disable PSG+ mode.
PSG PLAYaddr[, len]	Start hardware-mode AY playback.
PSG STOP	Stop AY playback.

A BASIC fragment that plays a tone on channel A:

```
10 POKE &H000F0800, 1          : REM AUDIO_CTRL = on
20 PSG MIXER &H38               : REM tones on, noise off
30 PSG 0, 113, 15               : REM divider 113, level 15
```

The exact divider for a desired frequency depends on the chip clock chosen; see §13.2.1.

13.6 Putting it together

To use the PSG: route tone or noise to each channel through the mixer, set each channel's divider and level, and optionally trigger the envelope by writing register 13. For existing chip-tune files, point `PSG_PLAY_PTR/LEN` at the register-frame stream and write 1 to `PSG_PLAY_CTRL`.

The next chapter covers the SN76489, the PSG's three-voice cousin from Texas Instruments.